9's complement of 864: $\begin{array}{r} 999 \\ -864 \\ \hline 135 \end{array}$ $\begin{array}{r} 753 \\ +135 \\ \hline 888 \end{array}$ 9's complement of 888: $\begin{array}{r} 999 \\ -888 \\ \hline 111 \end{array}$

Answer: −111

d) 20 − 1000 M = 0020 N = 1000

10's complement of N: $\begin{array}{r} 1000 \\ -1000 \\ \hline 0000 \end{array}$

$\begin{array}{r} 0020 \\ +0 \\ \hline 0020 \end{array}$

10's complement of 0020: $\begin{array}{r} 1000 \\ -0020 \\ \hline 980 \end{array}$

Answer: −980

9's complement of N: $\begin{array}{r} 9999 \\ -1000 \\ \hline 8999 \end{array}$ $\begin{array}{r} 0020 \\ 8999 \\ \hline 9019 \end{array}$ 9's complement of 9019: $\begin{array}{r} 9999 \\ -9019 \\ \hline 980 \end{array}$

Answer: −980

1-12: Perform the subtraction with the following binary numbers using (1) 2's complement and (2) 1's complement. Check the answer by straight subtraction.

(a) 11010 − 1101 M = 11010 N = 01101

2's complement of 01101: $\begin{array}{r} 10010 \\ +1 \\ \hline 10011 \end{array}$

$\begin{array}{r} 11010 \\ +10011 \\ \hline 01101 \end{array}$ End carry: 1

1's complement of 01101: 10010

$\begin{array}{r} 11010 \\ +10010 \\ \hline 01100 \\ +1 \\ \hline 01101 \end{array}$

$\begin{array}{r} 11010 \\ -01101 \\ \hline 1101 \end{array}$

(b) 11010 − 10000 M = 11010 N = 10000

2's complement of 10000: $\begin{array}{r} 01111 \\ +1 \\ \hline 10000 \end{array}$

$\begin{array}{r} 11010 \\ +10000 \\ \hline 01010 \end{array}$ End carry: 1

1's complement of 10000: 01111

$\begin{array}{r} 11010 \\ 01111 \\ \hline 01001 \\ +1 \\ \hline 01010 \end{array}$ → End carry

$\begin{array}{r} 11010 \\ -10000 \\ \hline 01010 \end{array}$

(c) 10010 − 10011 M = 10010 N = 10011

2's complement of 10011: $\begin{array}{r} 01100 \\ +1 \\ \hline 01101 \end{array}$

$\begin{array}{r} 10010 \\ 01101 \\ \hline 11111 \end{array}$

2's complement of 11111: $\begin{array}{r} 00000 \\ 1 \\ \hline 00001 \end{array}$

Ans: −1